DATE : 16/7/26

Exp. NO: 1

Basic Networking Commands

1. arp
2. hostname

```
C:\Users\Tcs>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a               Displays current ARP entries by interrogating the current
                   protocol data.  If inet_addr is specified, the IP and Physical
                   addresses for only the specified computer are displayed.  If
                   more than one network interface uses ARP, entries for each ARP
                   table are displayed.
  -g               Same as -a.
  -v               Displays current ARP entries in verbose mode.  All invalid
                   entries and entries on the loop-back interface will be shown.
  inet_addr        Specifies an internet address.
  -N if_addr       Displays the ARP entries for the network interface specified
                   by if_addr.
  -d               Deletes the host specified by inet_addr. inet_addr may be
                   wildcarded with * to delete all hosts.
  -s               Adds the host and associates the Internet address inet_addr
                   with the Physical address eth_addr.  The Physical address is
                   given as 6 hexadecimal bytes separated by hyphens. The entry
                   is permanent.
  eth_addr         Specifies a physical address.
  if_addr          If present, this specifies the Internet address of the
                   interface whose address translation table should be modified.
                   If not present, the first applicable interface will be used.

Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.

C:\Users\Tcs>hostname
DESKTOP-G0LVE4J
```

3. ipconfig

```
C:\Users\Tcs>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b0c9:acc6:6356:f35e%10
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::7aee:10f6:c7f9:b79c%12
   IPv4 Address. . . . . . . . . . . : 172.16.10.101
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1
```

4 nbtstat

```
C:\Users\Tcs>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

5. netstat , nslookup, pathping

```
C:\Users\Tcs>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49675        DESKTOP-G0LVE4J:49676  ESTABLISHED
  TCP    127.0.0.1:49676        DESKTOP-G0LVE4J:49675  ESTABLISHED
  TCP    172.16.10.101:60865    4.213.25.240:https     ESTABLISHED
  TCP    172.16.10.101:60918    [redacted]             CLOSE_WAIT
  TCP    172.16.10.101:60919    [redacted]             ESTABLISHED
  TCP    172.16.10.101:60923    13.107.213.254:https   CLOSE_WAIT
  TCP    172.16.10.101:60924    150.171.22.254:https   CLOSE_WAIT
  TCP    172.16.10.101:60925    13.107.246.58:https    CLOSE_WAIT
  TCP    172.16.10.101:60926    204.79.197.222:https   CLOSE_WAIT
  TCP    172.16.10.101:60928    4.213.25.241:https     ESTABLISHED

C:\Users\Tcs>nslookup
Default Server:  UnKnown
Address:  172.16.8.1

C:\Users\Tcs>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list      Loose source route along host-list.
    -h maximum_hops   Maximum number of hops to search for target.
    -i address        Use the specified source address.
    -n                Do not resolve addresses to hostnames.
    -p period         Wait period milliseconds between pings.
    -q num_queries    Number of queries per hop.
    -w timeout        Wait timeout milliseconds for each reply.
    -4                Force using IPv4.
    -6                Force using IPv6.
```

6 ping

```
C:\Users\Tcs>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name
```

7 route

```
C:\Users\Tcs>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f            Clears the routing tables of all gateway entries.  If this is
                used in conjunction with one of the commands, the tables are
                cleared prior to running the command.

  -p            When used with the ADD command, makes a route persistent across
                boots of the system. By default, routes are not preserved
                when the system is restarted. Ignored for all other commands,
                which always affect the appropriate persistent routes.

  -4            Force using IPv4.

  -6            Force using IPv6.

  command       One of these:
                  PRINT     Prints  a route
                  ADD       Adds    a route
                  DELETE    Deletes a route
                  CHANGE    Modifies an existing route
  destination   Specifies the host.
  MASK          Specifies that the next parameter is the 'netmask' value.
  netmask       Specifies a subnet mask value for this route entry.
                If not specified, it defaults to 255.255.255.255.
  gateway       Specifies gateway.
  interface     the interface number for the specified route.
  METRIC        specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.
```

```
If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The * matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, *224*.

Pattern match is only allowed in PRINT command.
Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT -6
    > route PRINT 157*            .... Only prints those matching 157*

    > route ADD 157.0.0.0 MASK 255.0.0.0  157.55.80.1 METRIC 3 IF 2
            destination^      ^mask      ^gateway     metric^    ^
                                                     Interface^
      If IF is not given, it tries to find the best interface for a given
      gateway.
    > route ADD 3ffe::/32 3ffe::1

    > route CHANGE 157.0.0.0 MASK 255.0.0.0 157.55.80.5 METRIC 2 IF 2

      CHANGE is used to modify gateway and/or metric only.

    > route DELETE 157.0.0.0
    > route DELETE 3ffe::/32
```

```
[subha@localhost ~]$ ip address show

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536
    inet 127.0.0.1/8 scope host lo

2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP>
    inet 10.0.2.15/24 brd 10.0.2.255

[subha@localhost ~]$ ip addr show
(Output same as above)

[subha@localhost ~]$ su -
Password:
[root@localhost ~]# ip address add 192.168.1.254/24 dev enp0s3

[root@localhost ~]# ip address del 192.168.1.254/24 dev enp0s3

[root@localhost ~]# ip link set enp0s3 up

[root@localhost ~]# ip link set enp0s3 down

[root@localhost ~]# ip link set enp0s3 promisc on

[root@localhost ~]# ip route add default via 192.168.1.254 dev enp0s3

[root@localhost ~]# ip route add 192.168.1.0/24 via 192.168.1.254

[root@localhost ~]# ip route add 192.168.1.0/24 dev enp0s3

[root@localhost ~]# ip route delete 192.168.1.0/24 via 192.168.1.254

[root@localhost ~]# ip route get 10.10.1.4
10.10.1.4 via 10.0.2.2 dev enp0s3 src 10.0.2.15

[root@localhost ~]#
```

9. nmcli connection

```
[root@localhost ~]# nmcli connection modify "Wired connection 1" ipv6.method manual ipv6.addresses

[root@localhost ~]# nmcli connection up Internal-LAN
Connection successfully activated.

[root@localhost ~]# ip address show enp1s0
2: enp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP
    link/ether 52:54:00:17:b8:b6 brd ff:ff:ff:ff:ff:ff
    inet 192.0.2.1/24 brd 192.0.2.255 scope global enp1s0
    inet6 2001:db8:1::fffe/64 scope global

[root@localhost ~]# ip route show default
default via 192.0.2.254 dev enp1s0 proto static metric 102

[root@localhost ~]# ip -6 route show default
default via 2001:db8:1::ffee dev enp1s0 proto static metric 102

[root@localhost ~]# cat /etc/resolv.conf
search example.com
nameserver 192.0.2.200
nameserver 2001:db8:1::ffbb

[root@localhost ~]# ping google.com
PING google.com (...) 56(84) bytes of data.
64 bytes from google.com: icmp_seq=1 ttl=117 time=18.2 ms
64 bytes from google.com: icmp_seq=2 ttl=117 time=17.9 ms
^C
--- google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss
[root@localhost ~]#
```

10. mtr, tcpdump, ifconfig

```
[subha@localhost ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0
        ether 08:00:27:12:34:56

[root@localhost ~]# mtr google.com
Start: 2026-07-24
HOST: localhost                Loss%   Snt   Last   Avg
  1.|-- 10.0.2.2               0.0%    10    0.5   0.6
  2.|-- 192.168.1.1            0.0%    10    2.1   2.3
  3.|-- google.com              0.0%    10   18.4  18.2

[root@localhost ~]# mtr -g google.com
(Displays numeric IP addresses)

[root@localhost ~]# mtr -b google.com
(Displays hostnames and IP addresses)

[root@localhost ~]# mtr -c 10 google.com
10 packets transmitted successfully.

[root@localhost ~]# dnf install -y tcpdump
Complete!

[root@localhost ~]# tcpdump -D
1.enp0s3
2.any
3.lo

[root@localhost ~]# tcpdump -i enp0s3
listening on enp0s3, link-type EN10MB

[root@localhost ~]# tcpdump -i enp0s3 -c 10
10 packets captured

[root@localhost ~]# tcpdump -i enp0s3 -c 10 host 8.8.8.8
10 packets captured

[root@localhost ~]# tcpdump -i enp0s3 src host 8.8.8.8
listening on enp0s3...

[root@localhost ~]# tcpdump -i enp0s3 dst host 8.8.8.8
listening on enp0s3...

[root@localhost ~]# tcpdump -i enp0s3 net 10.1.0.0/24
listening on enp0s3...

[root@localhost ~]# tcpdump -i enp0s3 port 53
listening on enp0s3...

[root@localhost ~]# tcpdump -i enp0s3 host 8.8.8.8 and port 53
listening on enp0s3...

[root@localhost ~]# tcpdump -i enp0s3 -c 10 host www.google.com and port 443
10 packets captured

[root@localhost ~]# tcpdump -i enp0s3 port not 53 and not 25
listening on enp0s3...

[subha@localhost ~]$ ping -c 4 google.com
64 bytes from google.com: icmp_seq=1 ttl=117 time=18.2 ms
64 bytes from google.com: icmp_seq=2 ttl=117 time=18.0 ms
64 bytes from google.com: icmp_seq=3 ttl=117 time=18.3 ms
64 bytes from google.com: icmp_seq=4 ttl=117 time=18.1 ms
--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss
```

Result:

Thus the windows and linux commands has been executed and verified successfully.

DATE: 30/7/26

EXP.NO: 2

Study of different types of Network Cables.

Aim:

To study different types of network cables.

a) Different types of network cable.

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable.
3. Coaxial cable
4. Fibre Optic Cable

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use
UTP	Category3	10 bps	Advantages * Cheaper in cost * Easy to install Disadvantages * More prone to EMI.	10 Base-T Ethernet
	Category5	Upto to 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category5e	1 Gbps		Fast Ethernet.
STP	Category6, 6a	10 Gbps	Advantages * Shielded * Faster than UTP Disadvantage * Expensive * Greater installation.	Gigabit Ethernet, 10GB Ethernet. Widely used in data centers.
SSTP	Category7	10 Gbps		

Cable Type	Category	Maximum data Transmission	Advantages/ Disadvantages	Application / Use
Coaxial cable	RG1-6 RG1-59 RG1-11	10-100 Mbps	Advantage * High bandwidth * Versatile Disadvantage * Cost * Size is bulky.	*Speed of signal is 500 m. * High speed internet.
Fibre optics cable	Single mode multimode	100 Gbps	Advantages * High speed * High security Disadvantage * Expensive.	Maximum distance of fibre optics

b) Make your own Ethernet Cross-over Cable / straight cable.

* Tools & parts needed
* Ethernet cabling CAT 5e is certified for gigabit support, but cat 5 cabling works as well, just over shorter distances.
* A crimping tool.
* Two RJ45 plugs.
* Optional two plug shields.

Procedure:

STEP-1: To start construction of the device, begin by threading shields onto the cable.

STEP-2 : Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round -area to complete this task.

STEP-3 : After, you will need to untangle the wires, there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom.

STEP-4 : Once the order is correct, bunch them together in a line and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing those into the RJ45 plug without messing up the order.

STEP-5 : Next, push the cable right in. The notch at the end of the plug needed to be just over the cable shielding & if it isn't, that means that you stripped off too much shielding.

STEP-6 : After the wires are securely sitting inside the plug, insert it into the crimping tool & push it.

STEP-7 : Lastly, repeat for the other end using diagram B using diagram A.

Result:

Thus, the different types of network cables is studied and implemented successfully.

DATE : 23/7/26

Exp NO: 3

Packet Tracer tool installation and User Interface

AIM:

To study the packet tracer tool installation and user interface overview.

Procedure:

1. From the network component box, click & drag and drop the below components:

   a) 4 Generic PCs and one HUB.

   b) 4 Generic PCs and one Switch.

2. Click ON connections

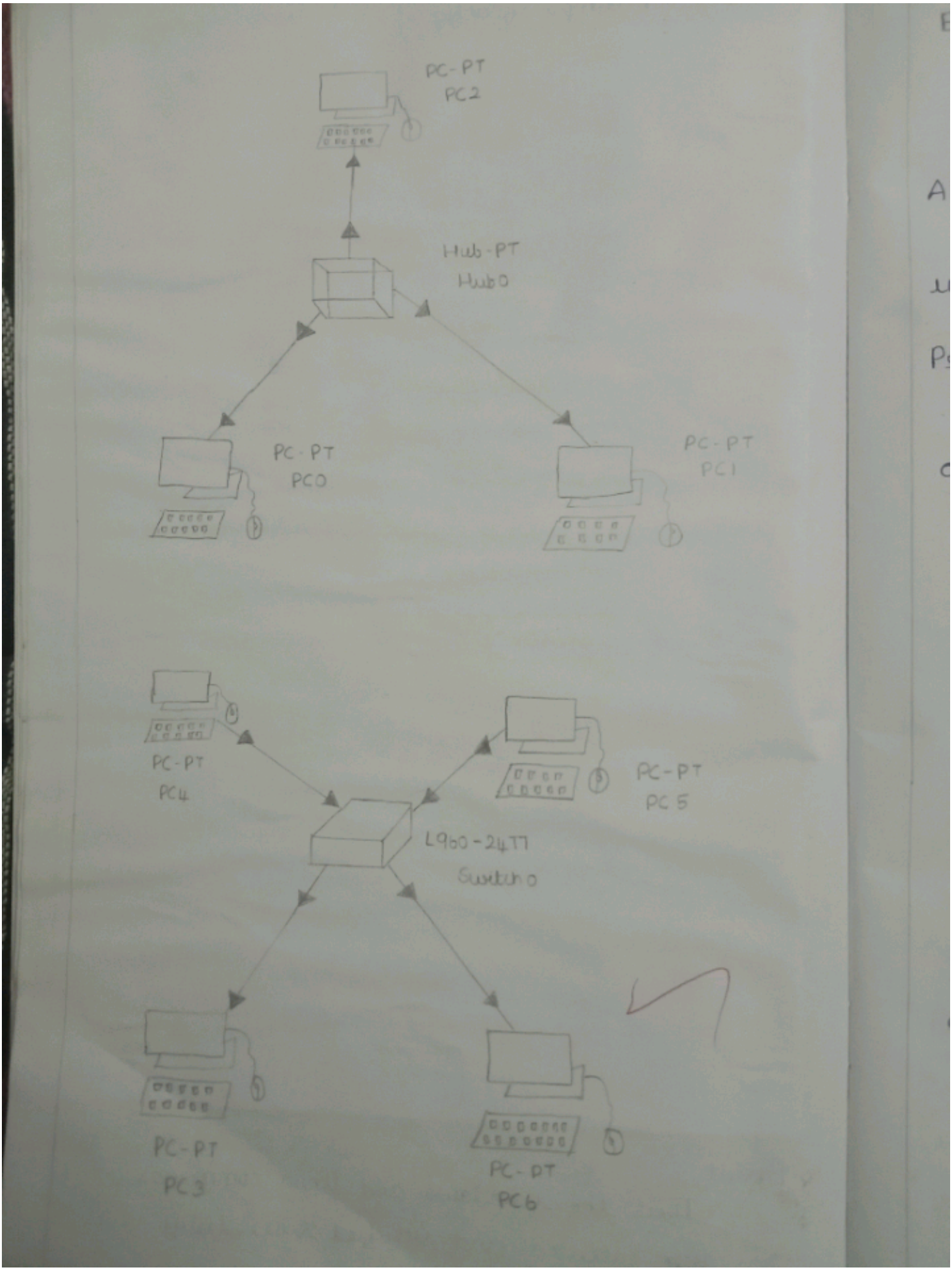
   a) Click on copper straight through cable.

   b) Select one of the PC and connect it to HUB using the cable. The link LED should glow in green, indicating that the link is up.

3. Click on the PCs connected to the HUB, go to the desktop tab, click on IP configuration and enter an IP address and subnet mask.

   a) Drag and Drop it on one of the PC & then drop it on another PC connected to the HUB.

4. Observe the flow of PDU from source PC to destination PC by selecting the realtime mode of simulation

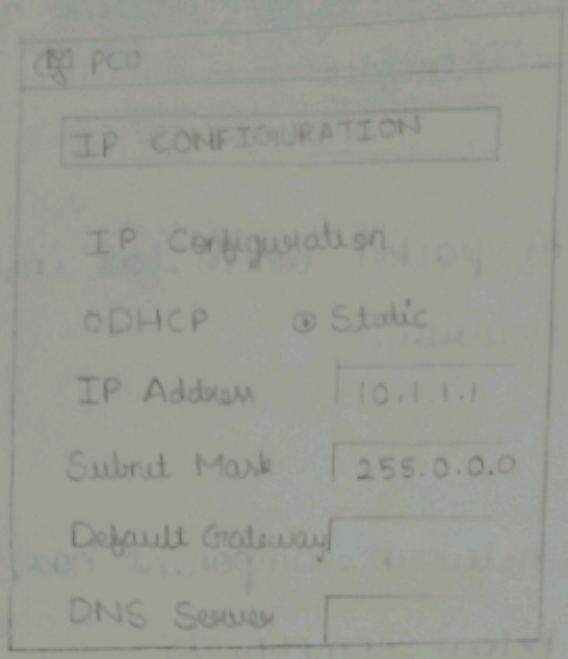

PC-PT
PC2
Hub-PT
Hub0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC4
PC-PT
PC5
2960-24TT
Switch0
PC-PT
PC3
PC-PT
PC6

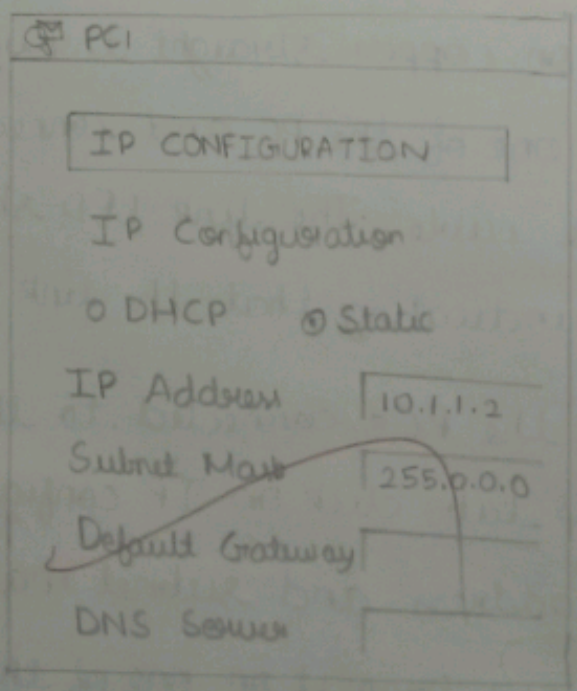
5. Repeat step 3 to step 5 for the PCs connected to the switch.

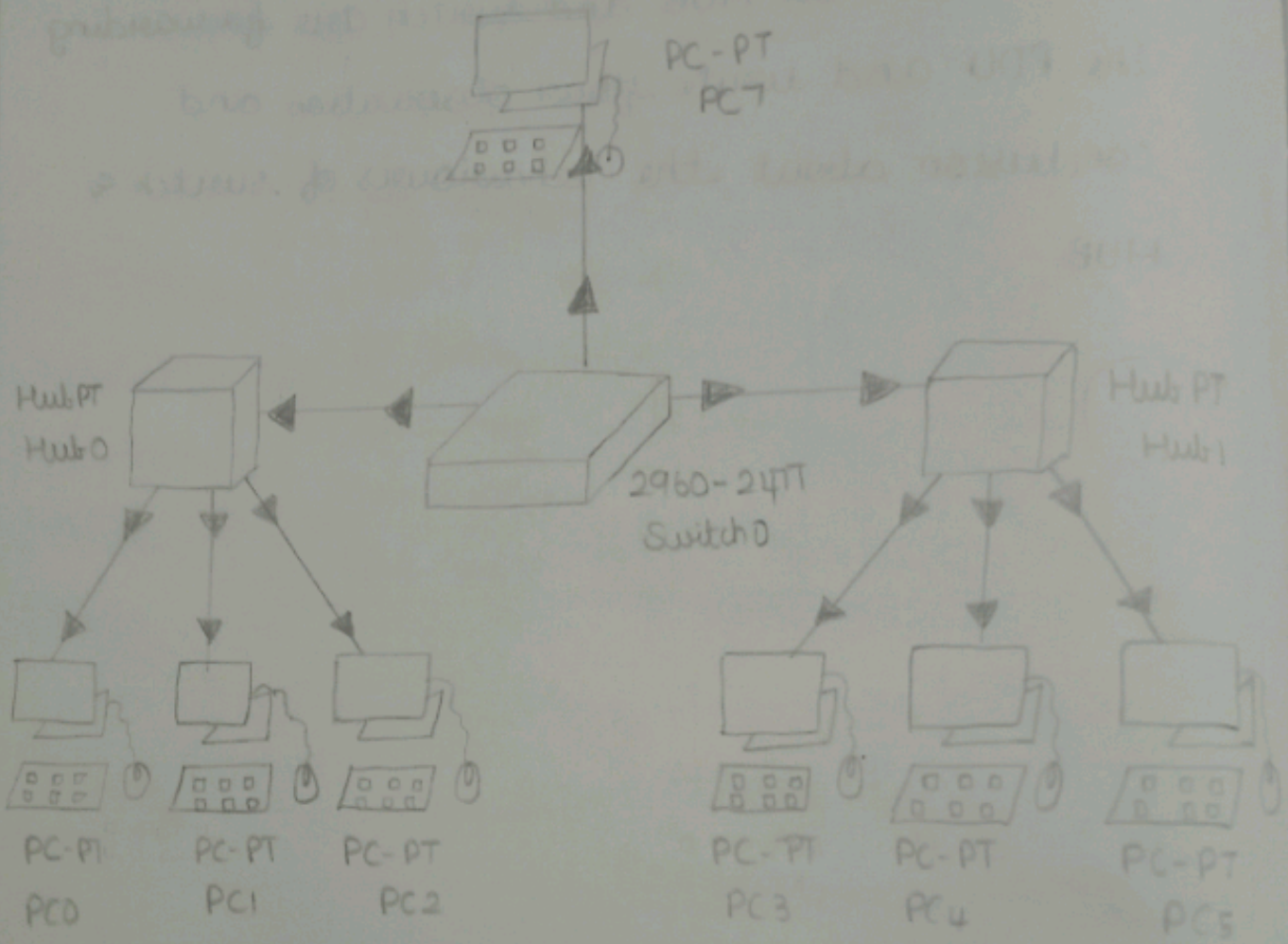
6. Observe how HUB and switch are forwarding the PDU and write your observation and conclusion about the behaviours of switch & HUB.

Result:

Thus, the study of packet tracer tool installation and user interface overview was successfully created and implemented.


PC0
IP CONFIGURATION
IP Configuration
DHCP
Static
IP Address
10.1.1.1
Subnet Mask
255.0.0.0
Default Gateway
DNS Server


PC1
IP CONFIGURATION
IP Configuration
DHCP
Static
IP Address
10.1.1.2
Subnet Mask
255.0.0.0
Default Gateway
DNS Server


PC-PT
PC7
Hub PT
Hub0
2960-24TT
Switch0
Hub PT
Hub1
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5

Working of Hub:

1. A hub receives data from one connected devices.
2. It broadcasts the data to all connected devices without checking the destination.
3. Every device receives the data, but only the intended device accepts it while others ignore it.

Working of Switch:

1. A switch receives data from the sendor device.
2. It checks the destination MAC address and identifies the correct port using its MAC address table.
3. It forwards the data only to the intended destination device, reducing unnecessary network traffic.

Working of switch & Hub:

1. Both are networking devices used to connect multiple devices within a LAN.
2. A hub sends data to all connected devices, whereas a switch sends data only to the intended device.
3. Switch produces better speed, security than Hub.

Result:

Thus, the study of packet tracer tool installation and user interface overview was successfully created and implemented.

DATE: 1/8/26

Exp NO: 4

Configuration of a LAN using a Switch and Ethernet cables.

Aim:

To setup and configure a LAN using a switch and Ethernet Cable.

Local Area Network (LAN)

It refers to a network that connects devices within a limited area, such as an office building, school or home.

It enables users to share resources, including data, printers and internet access.

Procedure:

1. Plan and design an appropriate network topology taking into account network requirements.
2. You can take 4 computers, 1 switch with 8, 16 or 24 ports which is sufficient for networks.
3. Connect your networks to network switch via an Ethernet cable, which is as simple as plugging one end of the Ethernet into your computer and the other in your network switch.

4. Assign IP address to your PC's

a) Log on to the client computer as owner.

b) Select Click network and internet connections.

c) Right click on Local Area Network.

5. Configure a network switch.

(i) Connect your computer to the switch. To access the switch's web interface, you will need to connect your computer to switch.

(ii) Log into the Web interface.

(iii) Configure basic settings.

(iv) Assign IP address as: 10.1.1.5, subnet mask 255.0.0.0

6. Check the connectivity between switch & other machine.

7. Select a folder → Properties → Share → Share it with everyone on the same LAN.

8. Try to access the shared folders from other computers.

Result:

Thus, the setup and configuration of LAN using switch and ethernet has been studied and implemented successfully.

DATE: 1/8/26

Exp NO: 5

Packet Capture Tool: WireShark

Aim:

To study the Packet capture tool - WireShark.

Packet Sniffer

1. Sniffs messages being sent/received by/from your computer
2. Store and display the contents.

Packet Sniffer Structure Diagnostic Tools.

1. Tcpdump

   Eg: tcpdump -enx host 10.129.41.2 -w exe3

2. Wireshark

   - wireshark -r exe3.out

Description:

WireShark

1. Wireshark a network analysis tool formerly known as Ethereal, captures packets in real time and display them in human-readable format.
2. It includes filters, colour, coding and other features that let you dig deep into network.

Getting Wireshark

1. Wireshark can be downloaded for Windows or MacOS from its Official Website.
2. For Linux, wireshark will be found in its package repositories.

Packet List Pane

1. It displays all the packets in the current capture file.
2. Each line in the packet list corresponds to one packet in the capture file.

Packet Details Pane

1. It shows the current packet in a more detailed form.
2. It also shows protocols and protocol fields of the packet selected in the Packet List.

Packet Bytes Pane

1. It shows the data of the computer in current packet in hexdump style.

Color Coding

1. Wireshark uses colors to help you identify the types of traffic at a glance.

2. By default, light purple - TCP traffic, light blue - UDP traffic & black - packets with error.

Procedure

1. Create a filter to display only TCP/UDP Packets, inspect packets and provide the flow graph.

2. Create a filter to display only ARP packets and inspect the packets.

3. Create a filter to display only DNS packets and provide the flow graph.

4. Create a filter to display only HTTP packets and inspect the packets.

5. Create a filter to display only IP/ICMP packets and inspect the packets.

6. Create a filter to display only DHCP packets and inspect the packets.

Result:

Thus, the packet capture tool - Wireshark has been implemented and executed successfully.

Date: 8/8/26

Exp NO: 6

Error Detection and Correction using Hamming Code

Aim:

To write a program to implement error detection and corresponding using the Hamming code.

Algorithm:

1. Start the program
2. Read the number of 4-bit data blocks (n) forming the input data stream.
3. For each 4-bit data block $D = d_1\ d_2\ d_3\ d_4$, perform Hamming Code.
   a) Arrange 7 bits with parity bits.
   b) Compute the parity bit using even parity.

$$P_1 = d_1 \text{ XOR } d_2 \text{ XOR } d_4$$

$$P_4 = d_2 \text{ XOR } d_3 \text{ XOR } d_4$$

4. Simulate transmission by introducing a single-bit error at a chosen bit position.
5. At the receiver, recompute the check bits

$$C_1 = P_1 \text{ XOR } d_1 \text{ XOR } d_2 \text{ XOR } d_4$$

$$C_4 = P_4 \text{ XOR } d_2 \text{ XOR } d_3 \text{ XOR } d_4$$

6. Compute Error Position : $C_4 C_2$

7. If Error Position = 0, no error has occured.

8. Extract the original 4 data bits from the corrected 7-bit code word.

9. Repeat steps 3 to 8 for all n blocks in the data stream.

10. Stop the program

PROGRAM:

```
def encode(d):
    """d: list of 4 bits [d1, d2, d3, d4] -> return 7-
       bit hamming code list Positions (1-indexed):
       1 = P12 = P23 = d14 = P4 5 = d2 6 = d3 7 = d4"""
    h = [0] * 7
    h[2] = d[0]
    h[4] = d[1]
    h[5] = d[2]
    h[6] = d[3]
    h[0] = h[2] ^ h[4] ^ h[6]
    h[1] = h[2] ^ h[5] ^ h[6]
    h[3] = h[4] ^ h[5] ^ h[6]
    return h

def detect_and_correct(h):
    c1 = h[0] ^ h[2] ^ h[4] ^ h[6]
```

```
c2 = h[1] ^ h[2] ^ h[5] ^ h[4]
c4 = h[3] ^ h[4] ^ h[5] ^ h[6]

error_pos = c1 * c2 * 2 + c4 * 4

if error_pos != 0
    print (f" Error detected at bit position:
            {error_pos} -> Correcting....")
    h[error_pos - 1] ^= 1

else:
    print (" No error detected")

return error_pos

def to_str(bits):
    return "".join(str(b) for b in bits)

pos = int(input("Enter bit position to introduce
 error (1-7, 0 = no error): "))
if 1 <= pos <= 7:
    h[pos-1] ^= 1
    print(f" Transmitted Code (with error
     induced at pos{pos}): {to_str(h)}")
else:
    print(f" Transmitted code (no error induced):
        {to_str(h)}")
print(f" Received Code : {to_str(h)}")
print(" Checking for error...")
detect_and_correct(h)
```

O/P:

Enter number of 4-bit data blocks in the stream: 2

Block 1

Enter 4 data bits : 1011

Original data bits : 1011

Encoded Hamming code : 0110011

Enter bit position to introduce error : 3

Transmitted code : 0100011

Received code : 0100011

Error detected at bit position : 3

Corrected code : 0110011

Recovered data bits : 1011

Block 2

Enter 4 data bits : 0110

Original data bits : 0110

Encoded Hamming code : 1100110

Enter bit position to introduce error : 6

Transmitted code : 1100100

Received code : 1100100

Error detected at bit position : 6

Corrected code : 1100110

Recovered data bits : 0110

```
print(f"Corrected Code : {to_str(h)}")
ord = [h[2], h[4], h[5], h[6]]
print(f"Recovered Data Bits : {to_str(ord)}")

if __name__ == "__main__":
    main()
```

Result:

The program to implement error detection and correction using the Hamming Code concept was successfully written in Python, executed and verified using a sample input data stream.

Date: 13/8/26

Exp.NO: 7

Flow control at Data Link Layer using Sliding Window Protocol

Aim:

To write a program to implement flow control at the Data link layer using the Sliding Window Protocol, and to simulate the flow of frames from one node to another.

Algorithm:

1. Start the Program.
2. Read the total number of frames to be sent, the sender's window size and a frame number to simulate as lost during transmission.
3. Initialize base = 0 and nextSeq = 0
4. Repeat while base < N:
   a) Determine the current sender window.
   b) Send all frames with sequence numbers from nextSeq up to (base + W - 1).
   c) If the lost frame lies within the frames just sent:
      (i) The receiver discards the lost frame in order & sends a cumulative acknowledgement for it.

O/P:

Enter number of Frames : 5

Enter window size : 2

Frames sent:

Frame 1 sent

Frame 2 sent

ACK received

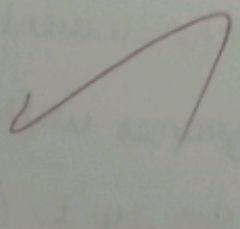
Frame 3 sent

Frame 4 sent

ACK received

Frame 5 sent

ACK received.



5. Repeat step 4 until all N frames have been transmitted and acknowledged.

6. Stop the program.

Program:

```
#include <stdio.h>
int main(){
    printf("Enter number of frames: ");
    scanf("%d", &n);
    printf("Enter window size: ");
    scanf("%d", &w);
    printf("\n Frames sent:\n");
    for(i=1; i<=n; i++){
        printf("Frame %d sent\n", i);
        if(i%w==0||i==n){
        printf("ACK received");
        }
        return 0;
    }
```

Result:

Thus, the sliding window protocol was successfully implemented in C.

Date

Exp No: 9

Implementation of Subnetting in Cisco Packet Tracer

Aim

To implement classless IP subnetting and configure a network using Cisco Packet Tracer.

Classless IP subnetting

* Subnetting divides a large network into smaller networks.
* Classless subnetting allows subnets masks other than the default Class A, B or C masks.

Network Topology

* Routers
* Switches
* PC's

Subnetting used

Given network - 192.168.1.0/24

To create subnets with sufficient host address, use:

/27 = 255.255.255.224

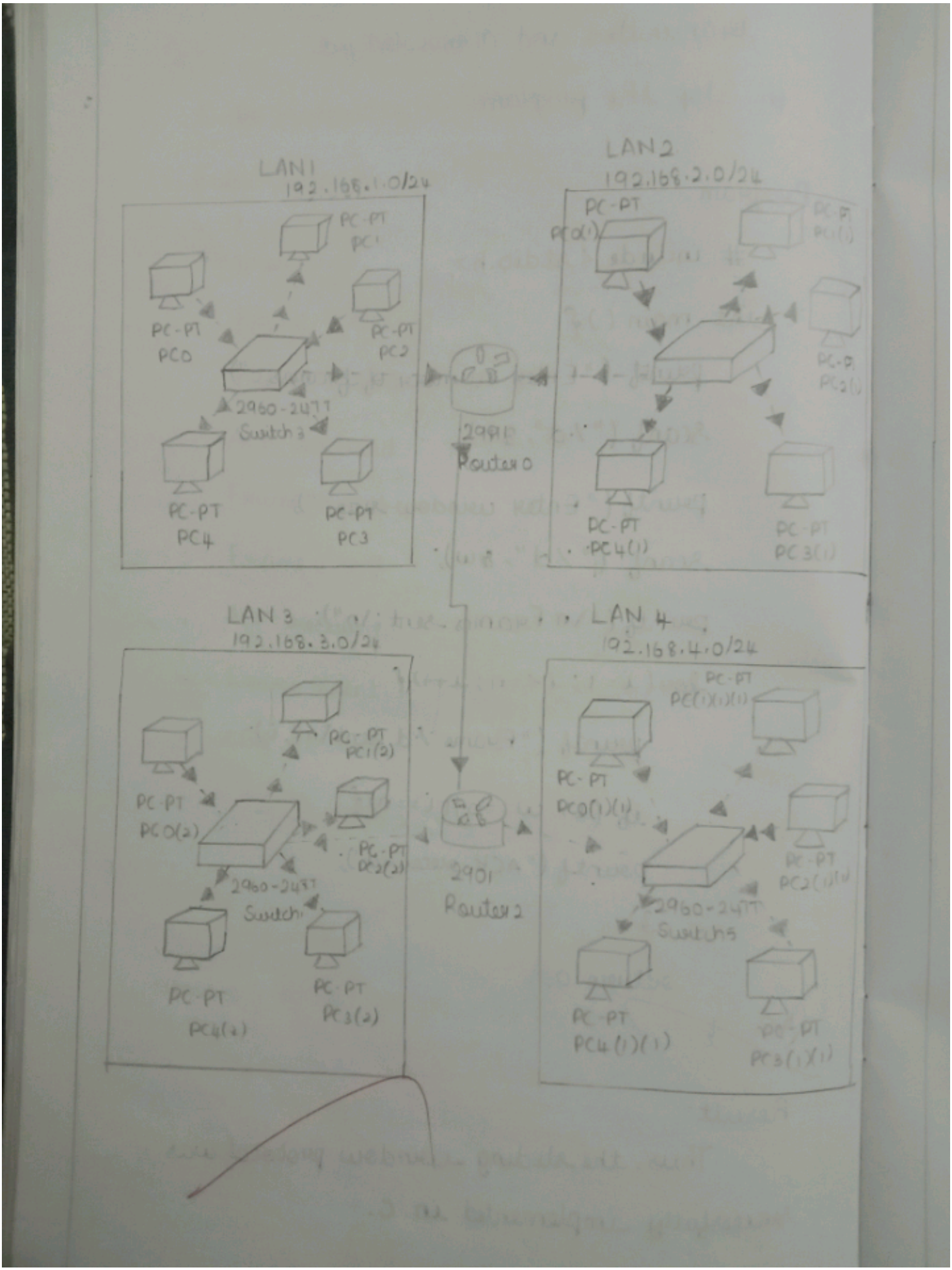
No: of subnets = 8

Usuable host addresses per subnet : 30

Block size : 32

Ex: 192.168.1.0/27

192.168.132 /27


LAN1
192.168.1.0/24
PC-PT
PC1
PC-PT
PC0
PC-PT
PC2
2960-24TT
Switch3
PC-PT
PC4
PC-PT
PC3
2911
Router0
LAN2
192.168.2.0/24
PC-PT
PC4(1)
PC-PT
PC3(1)
LAN3
192.168.3.0/24
PC-PT
PC1(2)
PC-PT
PC0(2)
PC-PT
PC2(2)
2960-24TT
Switch4
PC-PT
PC4(2)
PC-PT
PC3(2)
2901
Router2
LAN4
192.168.4.0/24
PC-PT
PC0(1)(1)
PC-PT
PC4(1)(1)
2960-24TT
Switch5

IP addressing

Network	Router IP	PCs
192.168.1.0/27	192.168.1.1	192.168.1.11 - 15
192.168.2.0/27	192.168.2.1	192.168.2.11 - 15
192.168.3.0/27	192.168.3.11	192.168.3.11 - 15
192.168.4.0/27	192.168.4.1	192.168.4.11 - 15

Procedure:

1. Open Cisco Packet Tracer and create a new network topology.
2. Add the required routers, switches & PC's.
3. Connect the devices using appropriate cables.
4. Divide the network 192.168.1.0/24 into smaller subnets using a /27 subnet mask.
5. Configure the router interfaces with suitable IP addresses and subnet masks.
6. Configure the switch port in access mode.
7. Assign IP addresses, subnet mask, and default gateway to each PC.
8. Enable router interface using the no shut down command.
9. Test connectivity between PC's and routers using the ping command.
10. Verify that all devices in the required

Router configuration:

enable
configure terminal
interface fastethernet 0/0
ip address <IP address> <subnet name>
no shutdown
exist.

For another interface

interface fast ethernet 1/0
ip address <IP address><subnet mask>
no shutdown
exit

interface GigabitEthernet 0/0
ip address <IP address><subnet mask>
no solution
exit.

Switch configuration

enable
configure terminal
interface FastEthernet 0/1
switchport mode access
exit

interface FastEthernet 0/2
switchport mode access
exit

Input

Network address : 192.168.1.0/24

Subnet Mask : 255.255.255.224 (/27)

Number of Subnets : 8

Usable Hosts per Subnet : 30

Output:

C:\> ping 192.168.1.12

Reply from 192.168.1.12: bytes = 32 time<1ms

Reply from 192.168.1.12: bytes = 32 time<1ms

Packets: Sent = 4, Received = 4, Lost = 0.

PC configuration

IP address

Subnet mask: 255.255.255.224

Default Gateway : Router's IP address for that subnet

DNS server if required.

Testing

ping < destination IP >

<1ms

<1ms

Result:

Thus, subnetting was successfully implemented and tested in Cisco Packet Tracer and connectivity between the devices was verified using the ping command.

Date

EXP NO: 8a)

NMap live host Discovery

Aim:

To discover live hosts in a network using Nmap with different host-discovery techniques such as ARP, ICMP, TCP & UDP scans on the TryHackMe platform.

Nmap

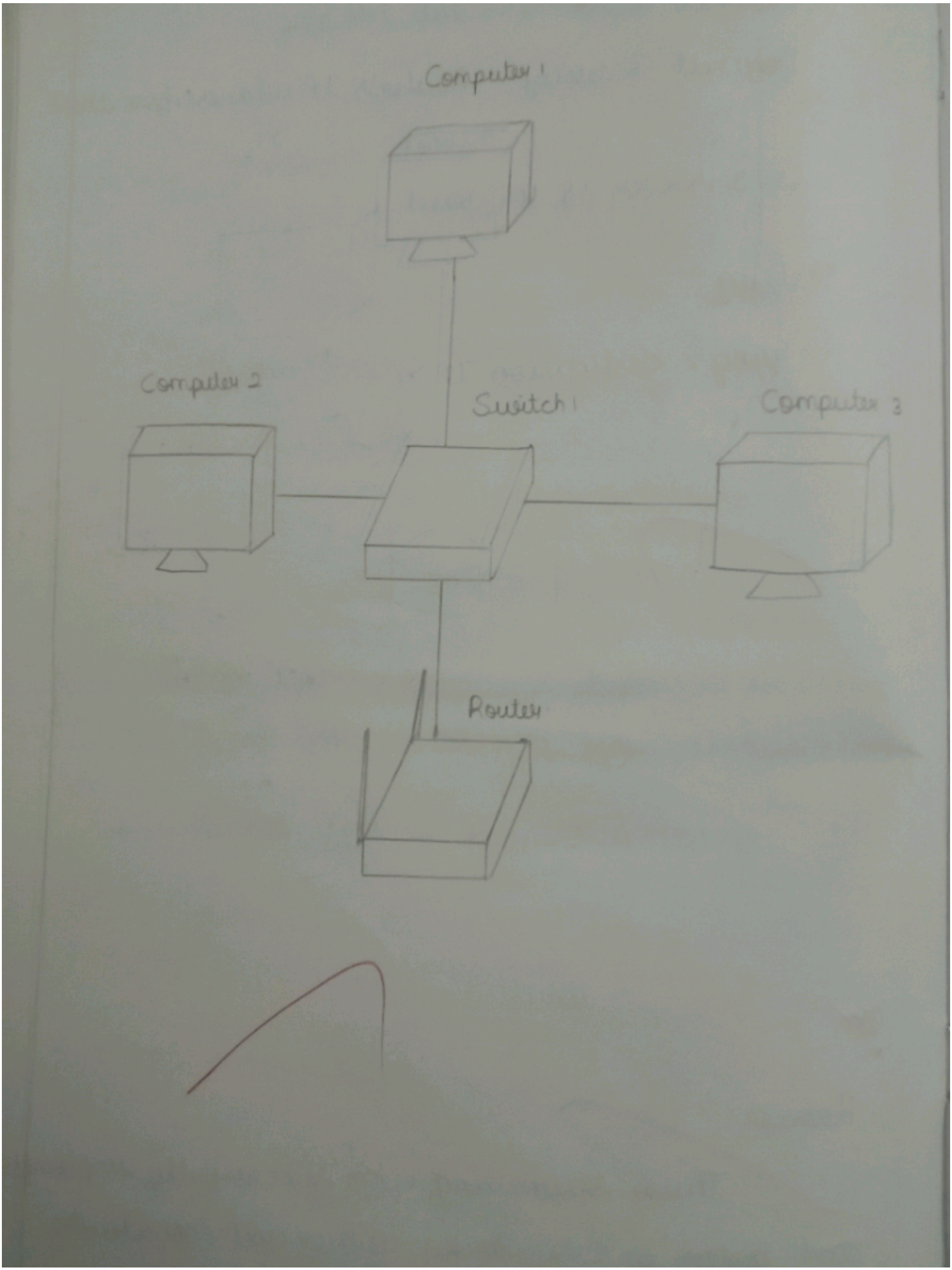
Network Mapper is a free and open source network scanning tool used to discover devices and services on a network.
It was developed by Gordon Lyon and was released in 1997

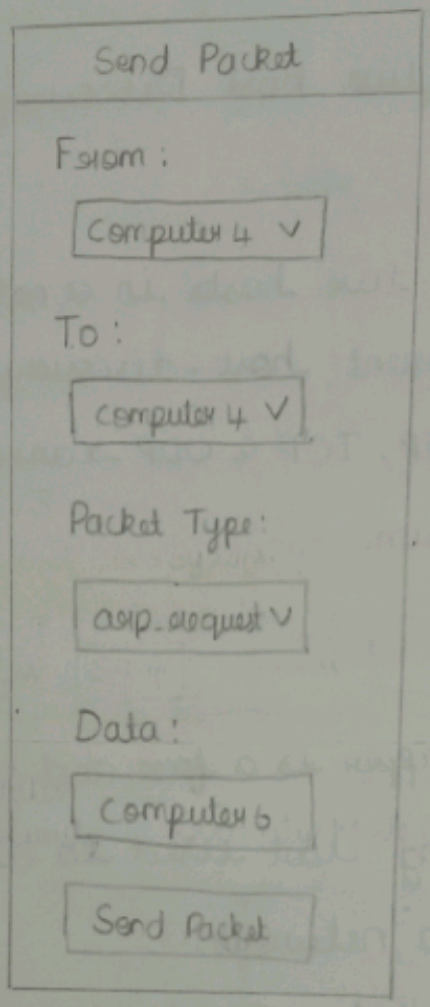
Subnetwork

A subnetwork is a logical part of a larger network. Each subnet has its own IP address range and is generally connected to other network through a router.

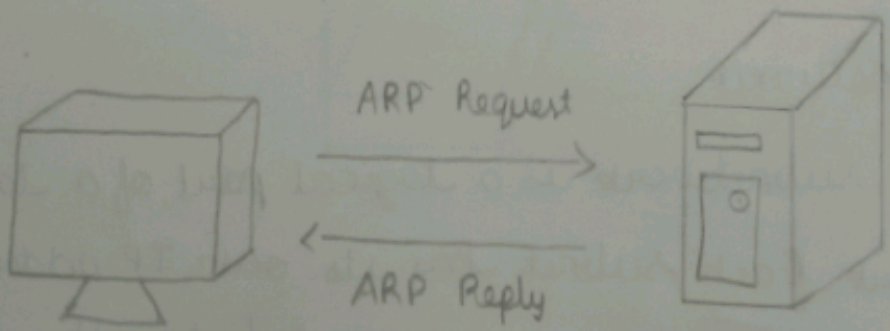
Live Host Discovery

Before scanning port, Nmap can determine which hosts are currently online. This avoids wasting time scanning systems that are offline.


Computer 1
Computer 2
Switch
Computer 3
Router



nmap -PR -Sn TARGET



Computer 1

Case : Host is live

Nmap uses a different protocols depending on the network is :

1. ARP scan

Address Resolution Protocol requests are used to discover active hosts on the same subnet. A reply indicates that the host is available on the network.

2. ICMP scan

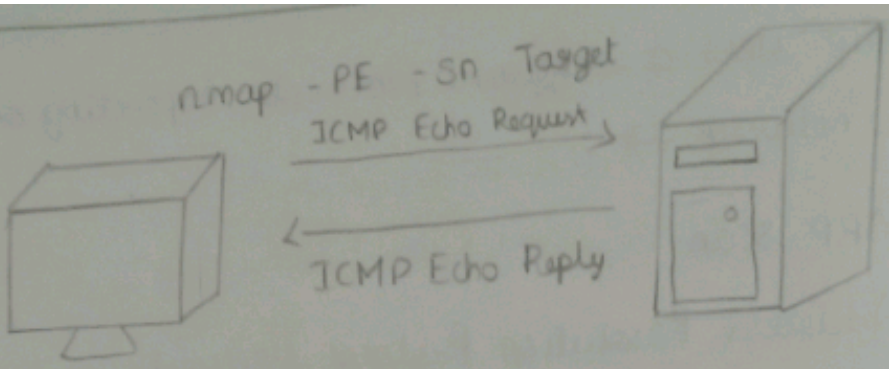
ICMP uses echo request and echo reply messages to check whether host is active. The -PE option is used for ICMP echo discovery.

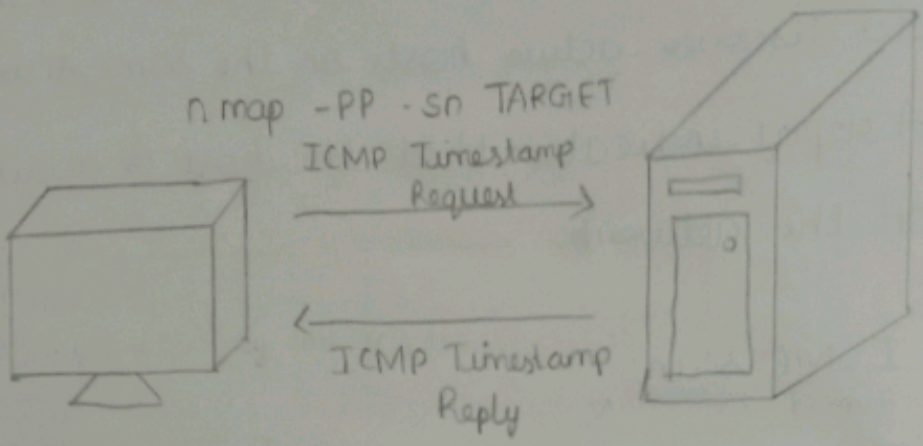
3. TCP syn Ping.

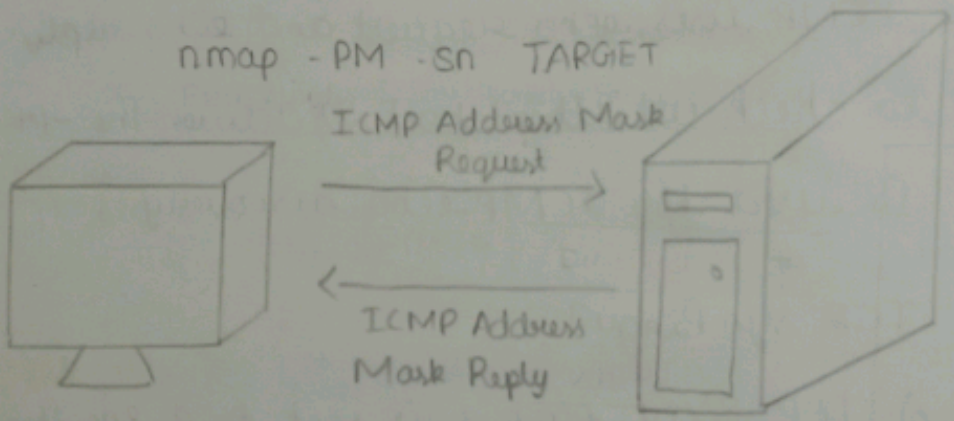
A TCP SYN packet is sent to a specified port. A response from the target indicates that the host is reachable. The -PS option is used for TCP SYN ping.

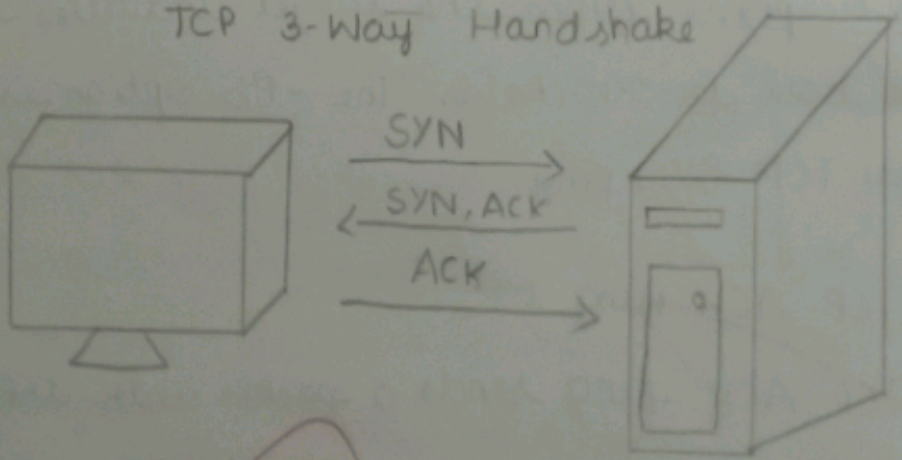
4. TCP ACK Ping

TCP ACK ping sends a packet with the ACK flag set. A response such as RST can indicate that the target hosts is active. The -PA option is used for this method.


nmap -PE -sn Target
ICMP Echo Request
ICMP Echo Reply


nmap -PP -sn TARGET
ICMP Timestamp Request
ICMP Timestamp Reply


nmap -PM -sn TARGET
ICMP Address Mask Request
ICMP Address Mask Reply


TCP 3-Way Handshake
SYN
SYN, ACK
ACK

UDP ping:

UDP packets can also be used for host discovery. A response such as an ICMP Port unreachable message can indicate that the target system is active.

Procedure:

1. Open the Nmap room in the TryHackme.
2. Study the given network topology & identify the target IP address.
3. Select the required Nmap host-discovery.
4. Perform ARP, ICMP, TCP SYN, TCP ACK where required.
5. Observe the packets and responses generated during the scans.
6. Use appropriate Nmap options such as -Sn, -PR, -PE and -PS.
7. Record the hosts that respond to the scans
8. Verify the result & identify active hosts.

Result:

Thus, the live hosts in the given network were successfully identified using Nmap. ARP, ICMP techniques were studied and used for host discovery.

Exp No: 8b)

Date

Configuration of wireless lan using Cisco packet Tracer

Aim:

To design a toplogy with three PC's connected from Linksys wireless Routers in Cisco Packet Tracer.

Perform following configuration:

Configure static IP on PC and wireless Router

Set SSID to Mother Network

Set IP address of router to 192.168.0.1

PC0 to 192.168.0.2, PC1 to 192.168.0.3 and PC2 to 192.168.0.4

Secure your network by configuring WAP key on router.

Connect PC by using WAP Key.

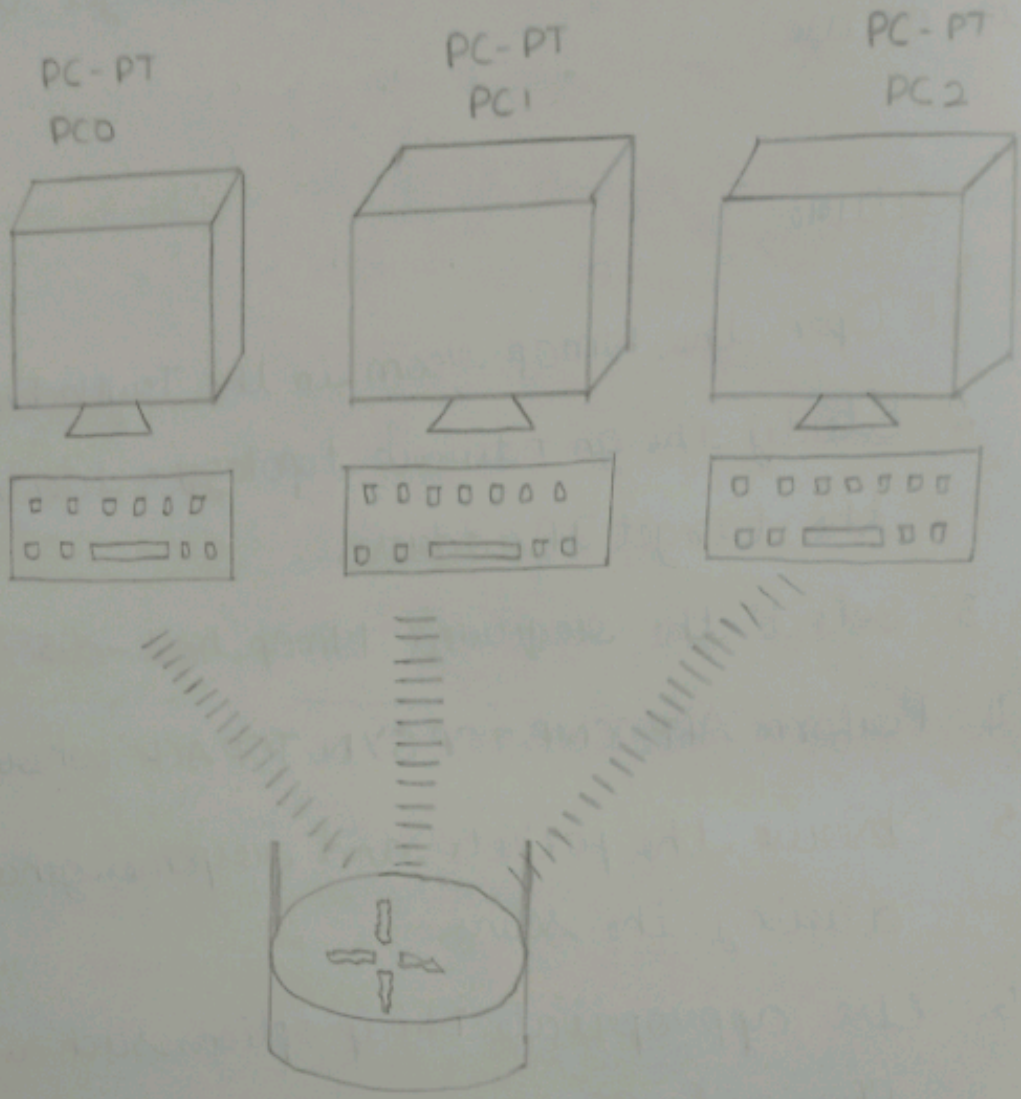
Procedure:

Step1: Click on Wireless router.

Select Administration tab from top Menu, set username and password to Admin and click on save setting.

Next click on wireless tab and set default SSID to mother network.

Select wireless security and change security mode to WEP.


PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Linksys - WRT300N
Wireless Router0

LINKSYS
A Division of Cisco Systems, Inc

Wireless | Setup | Wireless | Security | Access Restrictions | Applications & Gaming
Wireless Security | Wireless MAC Filter

Wireless Security

Security Mode : Disabled

Disabled
WEP

LINKSYS
A Division of Cisco Systems, Inc

Wireless | Setup | Wireless | Security | Access Restrictions | Applications & Gaming
Wireless Security | Wireless MAC Filter

Wireless Security

Security Mode : WEP

Encryption : 40/64-Bit (10 Hex digit)

Passphrase : Generate

Key 1 : 0123456789

Step 2

Set key1 to 0123456789

Again go in end of page and click on save setting

Now we have completed all given task on wireless router

PC	IP	Subnet mask	Default gateway
PC0	192.168.0.2		192.168.0.1
PC1	192.168.0.3		192.168.0.1
PC_2	192.168.0.4		192.168.0.1

Step 3 :

Click on connect tab & click refresh button.

The wireless device access mother network on CH6 and signal strength is 100%.

Select on connect button to connect mother notebook.

Result :

Thus, a wireless LAN with three PC'S was successfully configured using a linksys wireless router.

EXP. NO: 10.a)

Date:

Internetworking with Routers in Cisco Packet Tracer

Aim:

To design and configure a simple internetwork using one router and two PC's in Cisco Packet Tracer and verify connectivity between the PC's.

Network configuration

Device	Interface	IP Address
Router1	FastEthernet 0/0	192.168.10.1
Router1	FastEthernet 0/1	192.168.20.1
PC0	FastEthernet 0	192.168.10.2
PC1	FastEthernet 0	192.168.20.2

Procedure:

Step 1: Configure Router1

Open Router1 → CLI and enter

Router > enable

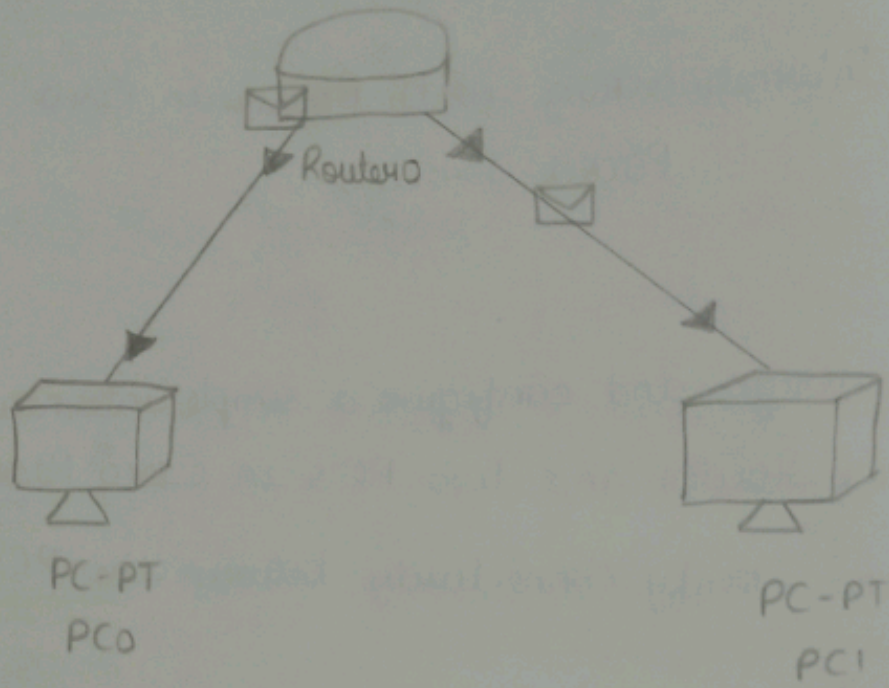
Router# configure terminal

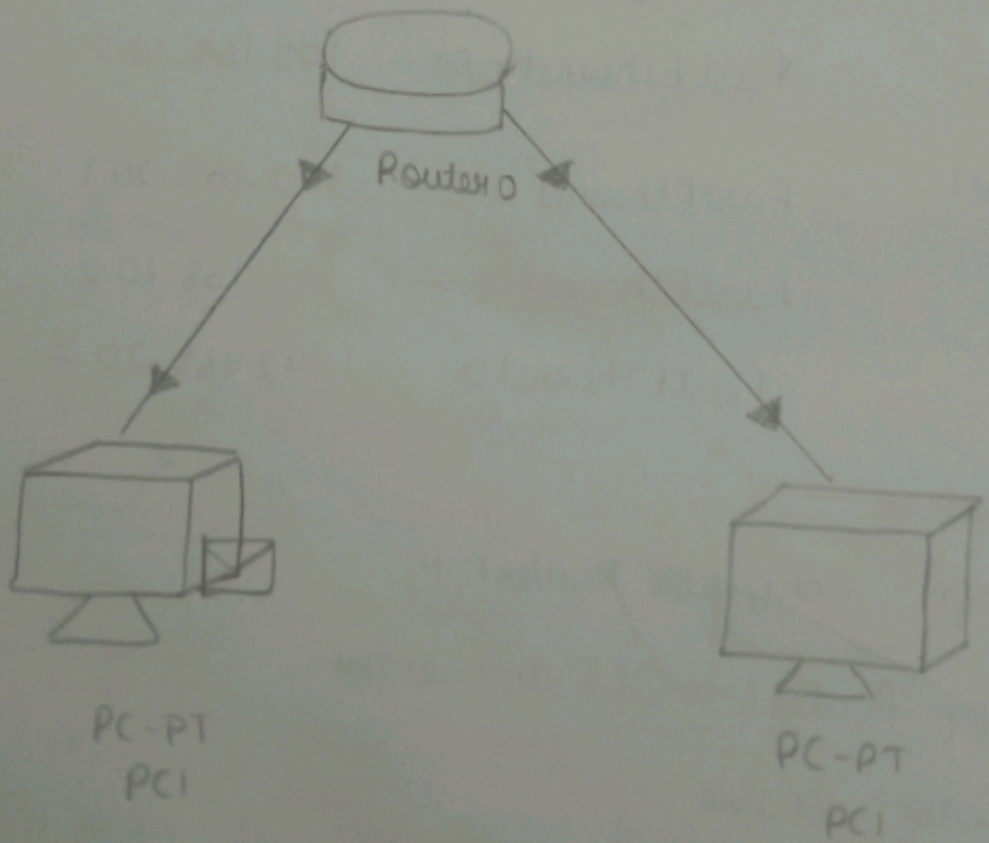
Router (config)

Router (config-if)

Router (config-if)

Acknowledgement from PC1 to PC0





Router (config-if) # interface FastEthernet 0/1
Router (config-if) # ip address 192.168.20.1 255.255.
Router (config-if) # no shutdown 255.0

Router (config-if)
Router (config)
Router #

Step 2: Configure PC0

Go to: PC0 → Desktop → IP Configuration

Enter:

IP Address: 192.168.10.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Step 3: Configure PC1

Go to: PC1 → Desktop → IP Configuration

Enter:

IP Address: 192.168.20.2

Subnet mask: 255.255.255.0

Default gateway: 192.168.20.1

Step 4: Connect the devices

Use copper straight-through cables from PC0 FastEthernet 0 to PC01 FastEthernet 0.

Step 5: Verify connectivity

From PC0 → Desktop → Command Prompt, use

ping 192.168.20.2

A successful reply indicates that PC0 can communicate with PC1 through Router1.

1. Select Add simple PDU.
2. Click PC0 as the source
3. Click PC1 as the destination.
4. Check the simulation window.
5. A successful PDU indicates connectivity.

Result:

A simple internetwork consisting of two PC's and one router was successfully designed and configured in the Cisco Packet Tracer.

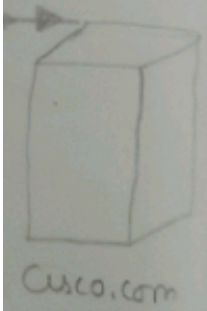
Exp NO: 10.b)

Date

Design and Configure an internetwork using wireless router, DHCP server & Internet Cloud.

Aim:

To design and configure an internetwork using a wireless router, DHCP server, laptop, PC, internet cloud, cable modem and cisco.com server and verify connectivity.

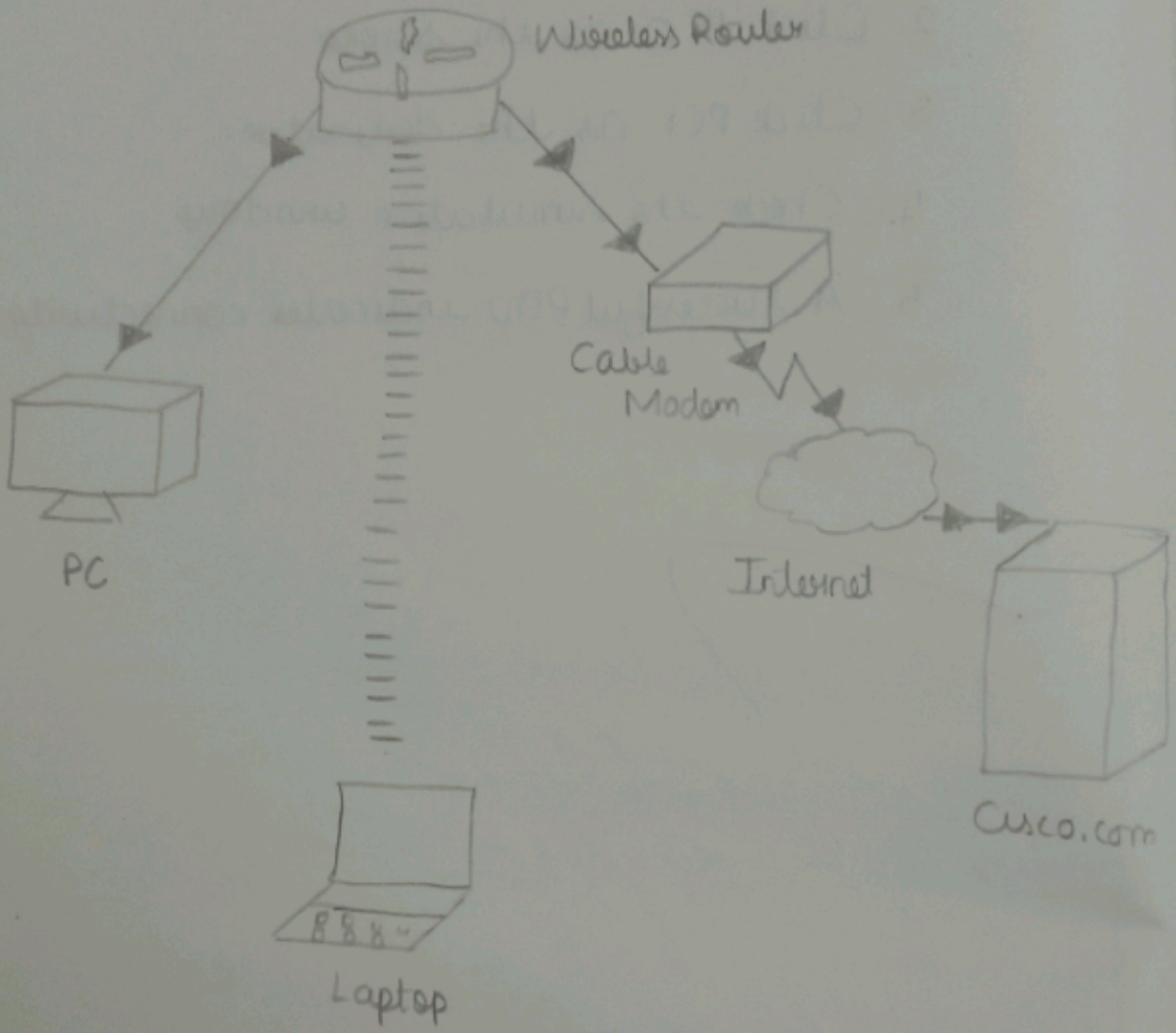


Objectives:

1. Build the work network topology.
2. Configure the wireless router and DHCP server.
3. Connect the laptop to the wireless network.
4. Configure the Cisco.com server as DHCP and ~~DNS~~ server.
5. Test connectivity between the PC & cisco.com server.

Addressing Table:

Device	Interface	IP Address
PC	Ethernet()	DHCP
Wireless Router	LAN	192.168.0.1
Wireless Router	Internet	DHCP


Wireless Router
Cable Modem
PC
Internet
Cisco.com
Laptop

Part-1 Build the network topology

Connections

From	To	Cable
PC FastEthernet()	Wireless Router Ethernet 1	Copper straight-through
Cable modem part 0	Internet cloud coaxial	Coaxial
Internet Cloud Ethernet	Cisco.com server FastEthernet()	Copper straight through

Part 2: Configure Network devices

Step 1: Configure wireless router

Open wireless router -> GUI -> Wireless

Set: Network Name (SSID): HomeNetwork

Step 2: Configure Laptop

Go to: Laptop -> Desktop -> PC wireless -> Connect

Step 3: Configure PC

Go to: PC -> Desktop -> IP Configuration -> DHCP

Step 4: Configure Internet Cloud

Open Internet Cloud -> Physical

Go to: Config -> Connection -> Cable

Wireless - N BroadBand Router

Firmware Version

Wireless - N BroadBand Router | WRT300N

Wireless | Setup | Wireless | Security | Access Restrictions | Application & Gaming | Administration

Basic Wireless Settings | Wireless Security | Guest Network

Basic Wireless Settings

Network Mode : Mixed

Network Name (SSID) : Home Network

Radio Band : Auto

Wide Channel : Auto

Standard Channel : 1 - 2.412 GHz

SSID Broadcast : ◉ Enabled ○ Disabled

Step 5: Configure Cisco.com server

A. configure DHCP service

Go to:

Cisco.com.server → Services → DHCP.

B. Configure DNS service

Go to: Services → DNS.

C. Configure Server Global settings

Go to: Config → settings.

D. Configure FastEthernet

Go to: Config → FastEthernet()

Part 3: Verify Connectivity

Step 1: Renew PC IP Address

Open: PC → Desktop → Command Prompt.

Step 2: Test Connectivity

Ping the server: ping 208.67.220.220

Result:

The internetwork was successfully designed and configured using a wireless router, DHCP server, DNS server, cable modem and Internet cloud.

Exp No: 11 a)

Date:

Static routing configuration using Cisco Packet Tracer

Aim:

To simulate and configure static routing using Cisco Packet tracer and verify the connectivity between different networks using main and backup routes.

Definitions:

Static Routing

It is a routing method in which routes are manually configured in the router's routing table by the network administrator.

Administrative Distance (AD)

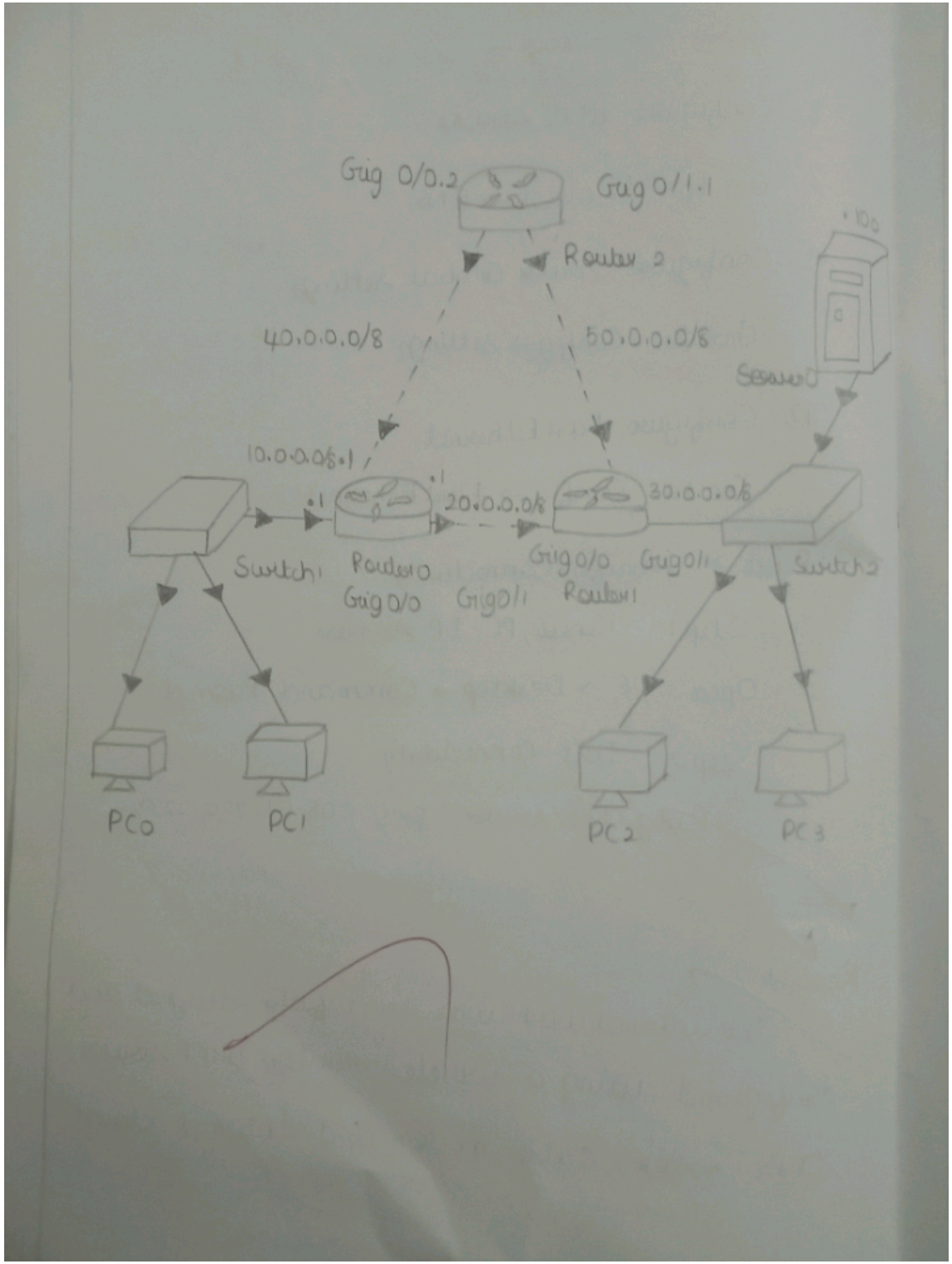
It is a value used by router to determine the preference of routes. A lower AD value is preferred over a higher AD.

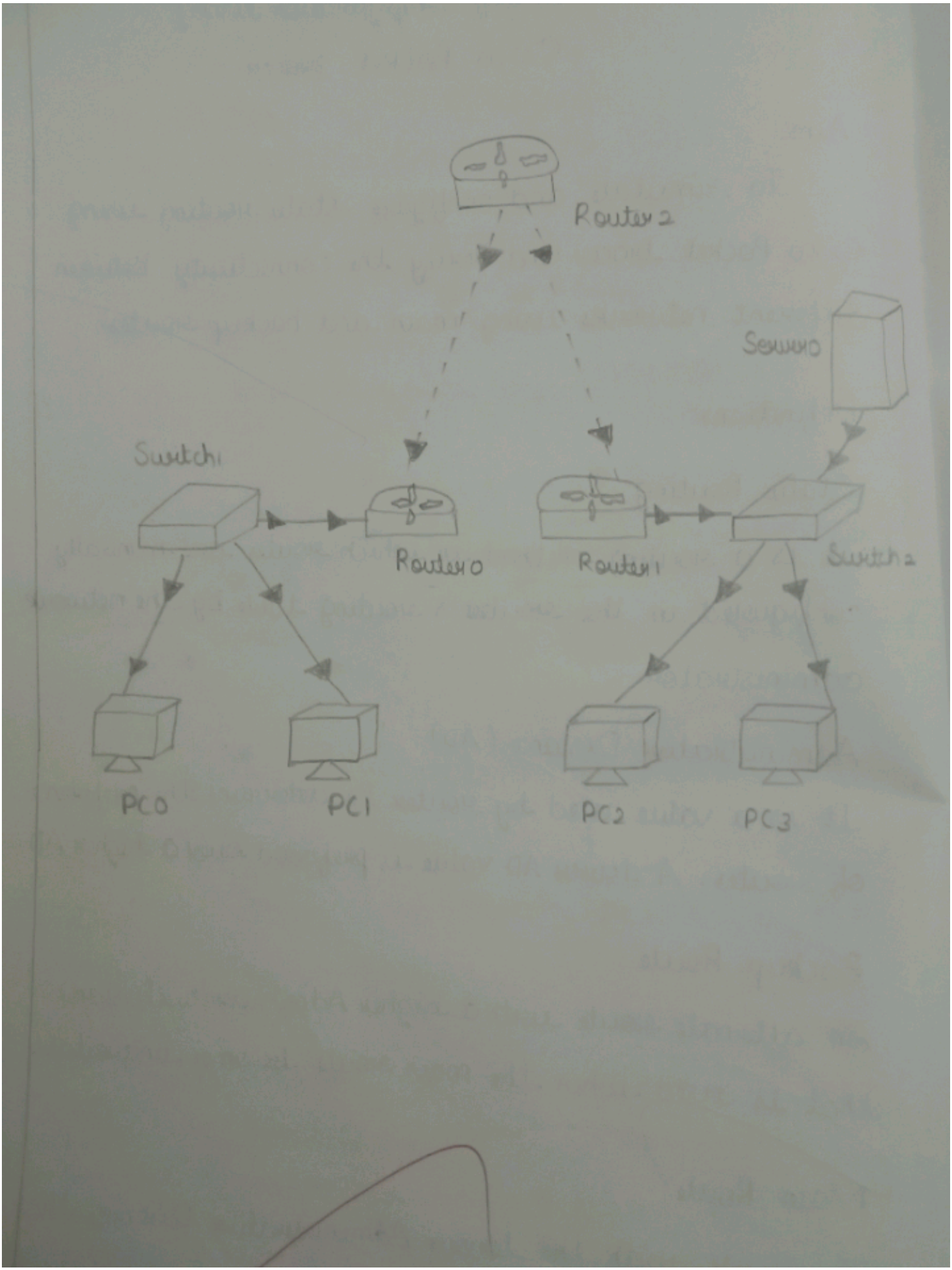
Backup Route

An alternate route with a higher Administrative Distance that is used when the main route becomes unavailable.

Main Route:

The route with the lower Administrative Distance that is used when the main route becomes unavailable.


Gig 0/0.2
Gig 0/1.1
Router 2
40.0.0.0/8
50.0.0.0/8
10.0.0.0/8
.1
.1
20.0.0.0/8
30.0.0.0/8
Switch1
Router0
Gig 0/0
Gig0/1
Gig 0/0
Gig0/1
Router1
Switch2
PC0
PC1
PC2
PC3

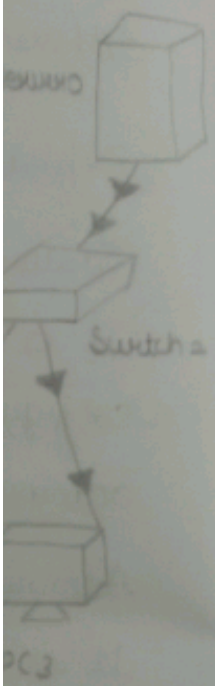

Router2
Server0
Switch1
Router0
Router1
Switch2
PC0
PC1
PC2
PC3

Procedure:

1. Open Cisco Packet Tracer.
2. Create the required network topology using routers, switches & PC's.
3. Connect the device using appropriate cables.
4. Configure the IP addresses and subnet masks on all router interfaces and PC's.
5. Open the CLI of Route() and enter privileged mode using: enable.
6. Enter configuration mode: configure terminal
7. Configure the required static routes using:

   ip route <destination-network> <subnet-mask>
   <next-hop-address> <AD>

8. Configure main and backup router by assigning lower and higher Administrative Distance values respectively.
9. Similarly, configure the required static routes on Router 1 and Router 2
10. Verify the connectivity & again using ping & traceroute



Result:

Thus, the Static Routing configuration was successfully simulated using Cisco Packet Tracer, and the main & backup routes were configured and verified successfully

Exp No: 11.b)

Date:

Simulate of RIP using Cisco Packet Tracer.

Aim:

To simulate Routing information Protocol (RIP) using Cisco Packet Tracer and verify dynamic routing and automatic route selection between different networks.

Important Definitions:

RIP (Routing Information Protocol)

It is a dynamic routing protocol used by routers to exchange information about networks and automatically determine routes

DCE (Data Circuit - terminating Equipment)

DCE is the end of a serial connection that provides the clocking signal.

DTE (Data Terminal Equipment)

DTE is the end of a serial connection that receives the clocking signal.

Hop Count

It is the number of routers a packet must pass through to reach the destination.

RIP selects the route with the lowest hop count.

Procedure:

1. Open cisco packet Tracer

2. Create the network topology using three routers (Router0, Router1, Router2) and two PC's.

3. Connect the device using appropriate Ethernet and serial cables.

4. Configure the IP address of PC0 and PC1 according to the given IP configuration.

5. Configure the IP addresses of PC0 and PC1 of the router interfaces.

6. Use the no shutdown command to activate the router interfaces.

7. Verify the DCE/DTE side using

show controllers serial 0/0/0

8. Verify RIP on Router0 using:

router rip

network 10.0.0.0

network 192.168.1.252

network 192.168.1.248.

9. Configure RIP on Router1 using:

router rip

network 192.168.1.244

network 192.168.1.248

10. Configure RIP on Router 2 and advertise its directly connected networks.

11. RIP automatically exchanges routing information between the routers.

12. Test connectivity between PC0 and PC1 using the ping command.

13. Use the tracert command to identify the route followed by packets from PC0 to PC1.

14. The network contains two possible routes between PC0 and PC1

15. Test the connectivity again using the ping command.

Result:

Thus, RIP was successfully simulated using Cisco Packet Tracer. Dynamic working between the routers was verified using ping & tracert & RIP successfully selected an alternate route when the primary route failed.

Exp No: 12a)

Date:

Echo client-server using TCP sockets

Aim:

To implement an echo client-server application using TCP sockets.

Algorithm:

Server

1. Create a TCP socket.
2. Bind the it to an IP address and port.
3. Listen for client connections.
4. Accept the client connection.
5. Receive the message and send it back.
6. Close the connection.

Client

1. Create a TCP socket.
2. Connect to the server.
3. Send a message
4. Receive a echoed message
5. Display the message
6. Close the connection.

Input:

Enter message : Hello Server

Output

Server : Hello Server

Code:

Server

```
import socket
s = socket.socket()
s.blind("localhost", 5000)
s.listen(1)
c, addr = s.accept()
msg = c.recv(1024).decode()
c.send(msg.encode())
c.close()
s.close()
```

Client

```
import socket
s = socket.socket()
s.connect(("localhost", 5000))
msg = input("Enter message:")
s.send(msg.encode())
print("Echo:", s.recv(1024).decode())
s.close()
```

Result:

Thus, the echo client-server application was successfully implemented using TCP sockets

Exp No: 12b)

Date:

Chat client-server using TCP sockets

Aim:

To implement a simple chat client-server application using TCP sockets.

Algorithm:

Server

1. Create a TCP socket.
2. Bind and listen for a client
3. Accept the connection.
4. Receive a message from the client.
5. Send a reply to the client
6. Close the connection

Client

1. Create a TCP socket.
2. Connect to the server.
3. Send a message
4. Receive the server's reply.
5. Display the reply.
6. Close the connection

Input

Client : Hello

Server : Hi

Output

Client : Hello

Server : Hi

Code - server

```
import socket
S = socket.socket()
S.bind ("localhost", 5000)
S.listen(1)
c, addr = S.accept()
while True:
    msg = c.recv(1024).decode()
    if msg == "bye": break
    print ("Client", msg)
    C.send (input ("Server:").encode())
c.close()
S.close()
```

Code - client

```
import socket
S = socket.socket()
S = connect (("localhost", 5000))
while True:
    msg = input ("Client:")
    S.send(msg.encode())
    if msg == "bye": break
    print ("Server:", S.recv(1024.decode())
S.close()
```

Result:

Thus, the chat client server was successfully implemented using TCP socket.

Exp NO: 13

Date:

Ping Program using Socket

Aim:

To implement a simple ping program using socket programming to check the connectivity between the source and destination hosts.

Algorithm:

1. Start the program.
2. Get the destination IP address from the user.
3. Create a raw socket.
4. Construct an ICMP echo request packet.
5. Send the packet to the destination.
6. Wait for an ICMP echo reply.
7. If a reply is received, display Reply received.
8. If no reply is received within the time out, display request Timed out.
9. Stop the program.

Code:

```
import socket
import struct
import time
```

Input :

Enter destination IP address : 8.8.8.8

Output :

Pinging 8.8.8.8

Reply received from 8.8.8.8

```
host = input ("Enter IP address")
S = socket.socket (socket.AF_time ;socket.SOCK_RAW,
                   socket.IPPROTO_ICMP)
S.settlement (2)
packet = struct.pack ("! BBHHH", 8, 8, 8, 1, 1)
S.sendto (packet, (host, 0))
try:
    data, adda = s.recvfrom (1024)
    print ("Reply received from", adda [0])
except socket.timeout:
    print ("Request timed out")
S.close()
```

8.8

Result:

Thus, the ping program was successfully implemented using socket programming and the connectivity between the source and destination host was verified.

Exp. NO: 14

Date:

Packet Sniffing using Raw sockets

Aim:

To write a program using RAW sockets to implement packet sniffing.

Algorithm:

1. Create a raw socket.
2. Bind the socket to the network interface.
3. Recieve network packets using the RAW socket.
4. Extract the packet header information.
5. Display the source IP, destination IP and protocol.
6. Continue capturing packets and display the details.

Code:

```
import socket
import struct

S = socket.socket (socket.AF_INET, socket.SOCK_RAW
                   socket.IPROTO_IP)

print ("Packet Sniffer Started...")
```

Input:

Network packets transmitted through the network.

Output:

Packet sniffer started ....

Source IP : 192.168.1.10

Destination IP : 142.250.183.14

Protocol : 6

Source IP : 192.168.1.10

Destination IP : 142.250.72.14

Protocol : 17

```
while True:
    packet, addr = S.recvfrom(65565)
    ip_header = packet[0:20]
    iph = struct.unpack('!BBHHHBBH4S4S', ip_header)
    source_ip = socket.inet_ntoa(iph[8])
    destination_ip = socket.inet_ntoa(iph[9])
    protocol = iph[6]
    print("Source IP :", source_ip)
    print("Destination IP :", destination_ip)
    print("Protocol :", protocol)
    print("-"*40)
```

Result:

Thus, the program to implement packet sniffing using RAW sockets was executed successfully.

Exp. No : 15

Date :

Web log analysis using webalizer

Aim:

To analyze different types of web logs using the webalizer tool.

Procedure:

1. Run the Webalizer windows version.
2. Input the web log file downloaded from the web.
3. Press Run Webalizer.
4. Webalizer processes the log file and generates statistical reports.
5. View the generated report in a web browser.
6. Webalizer reads and processes the log file.
7. It analyses the requests and visitor information.

Logfile:

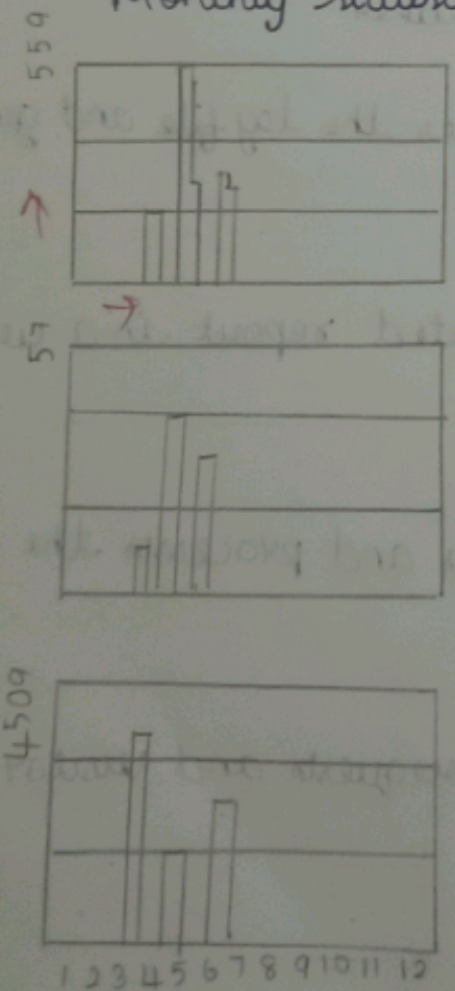
C:\Users\TCS\Downloads\access_log

Target Directory:

C:\Users\TCS\

Output:

Monthly statistics



8. The tool generates statical reports and graphs.
9. Open the generated report in a web browser.
10. Analyze the number of visits, pages and requests, bandwidth usage.

Result:-

Thus, the different types of web logs were successfully analyzed using the Webalizer tool.